

ENGS 28, Embedded Systems

Lab 1 Prelab

An embedded system engineer needs a solid understanding of basic electronics to effectively use sensors and actuators and interface them to a microcontroller. The purpose of this supplementary material is to review the basic principles of voltage and current from Physics 14, and then to introduce an electronic component, the light emitting diode (LED), that you will use in Lab 1. This is graded on completion, and you will be expected to be comfortable with this material.

You are encouraged to discuss these problems with a partner, making sure that you can explain the principles and solutions to each other, and answer each other's questions. Feel free to use your Physics 14 notes for review or consult a good online reference. Also, refer to the [ENGS 31 eBook](#) as an additional resource on electric circuits.

Useful Equations/Concepts:

1. Ohm's Law: $V = IR$, where V is the voltage *across* a circuit element, I is the current *through* the element and R is the resistance *of* the element.
2. Positive Current Convention: Current always flows from + to -.
3. Kirchhoff's Voltage Law: $\sum V_{loop} = 0V$. In words, the sum of the voltages around any loop is zero.
4. Kirchhoff's Current Law: $\sum I_{node} = 0A$. In words, the sum of the currents into a node is equal to the sum of the currents leaving the node.
5. Resistors in series: $R_{\text{effective}} = R_1 + R_2$.
6. Resistors in parallel: $R_{\text{effective}} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_{n-1}}}$. When $n=2$, $R_{\text{effective}} = \frac{R_1 R_2}{R_1 + R_2}$
7. Power dissipated in an element: $P = IV$ (W).

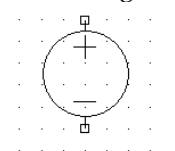
Circuit Schematic Symbols

1. A resistor:



A resistor impedes the flow of electric current.

2. A voltage source:



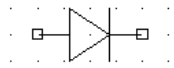
A voltage source is like a battery. An ideal voltage source produces a specified voltage and will source or sink as much current as needed.

3. Ground:



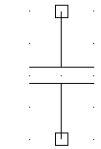
Ground is the 0-voltage reference for the system.

4. A diode:



Current flows in one direction only, from the anode (triangle) to the cathode (flat line). Note: A handy mnemonic is that the current flows in the direction of the arrow.

5. A capacitor:



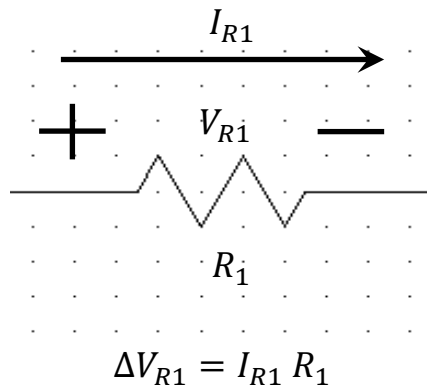
A capacitor stores electric charge. It blocks constant (direct) current and passes time-varying (alternating) current.

Common Terminology:

1. **Node:** An electrical connection between two or more circuit elements.
2. **Power or power rail:** The positive voltage used to supply current to the circuit. In our case, this will usually be positive 5 volts. It is often called Vcc or Vdd. Occasionally, there is more than one voltage source in a circuit. If there is any ambiguity about *which* supply is considered the power rail, be specific – “connect pin 8 to the power rail” becomes “connect pin 8 to the 5V pin on the Arduino’s power connector”.
3. **Ground or ground rail:** This is the zero-volt reference for our circuits. In general, we will not have any negative voltages in our designs, so this makes ground the most negative supply in our system. This is sometimes referred to as Vss or (rarely) Vee.
4. **The rails or the supply rails:** Collectively, this refers to the accepted power and ground rails in the system. As we become more familiar with constructing circuits, you may see an instruction to “connect the device to the rails”. This can be translated as “connect the power supply pin of the device to the 5v supply and connect the ground pin to the system’s ground node.”

Topic: Ohm's Law Review:

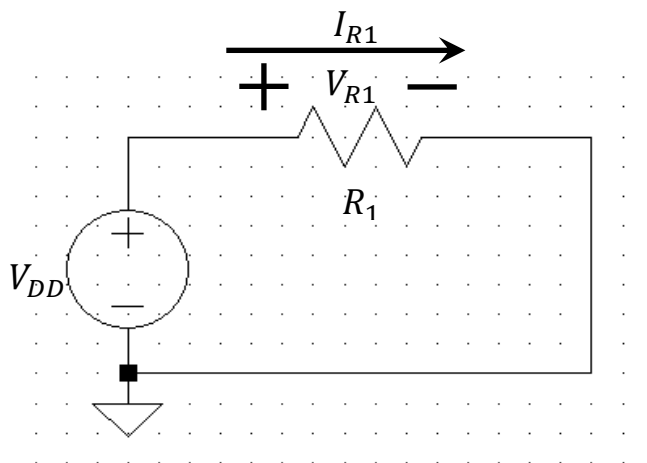
Ohm's Law relates the voltage drop across a resistive element: $V = IR$.



Ohm's Law is defined for a resistor in this drawing. Notice the relationship between the polarity (+/-) and the current. Although electrons are negatively charged, for historical reasons engineers follow the *positive current convention*, assuming current to **ALWAYS** flow from positive to negative. When analyzing a circuit with resistors, it doesn't matter which way you define your polarity, just keep current flowing consistently from + to -. The math will work itself out correctly if you do this.

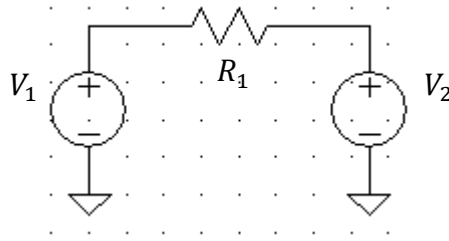
Problem 1: Ohm's Law:

In this problem the polarity is defined for you. In future problems, you'll need to define the polarity yourself. Solve for I_{R1} in terms of V_{DD} and R_1 , given $V_{DD} = 5V$, $R_1 = 220\Omega$. First write the solution algebraically, then substitute numerical values for the variables.



Problem 2: Ohm's Law Continued:

For the diagram below,



- Define polarities for I_{R1} and V_{R1} and label the diagram appropriately, showing the direction of current flow.
- Solve for I_{R1} in terms of V_1 , V_2 , and R_1 , where $V_1 = 5V$, $V_2 = 0.7V$, and $R_1 = 220 \Omega$. Write your solution algebraically, then substitute numerical values for the variables.
- If you define the polarity of V_{R1} in the opposite way, with the direction of current flow also reversed in order to flow from + to -, what happens to your calculation? Note that the electrons don't care about your notation, so your mathematical result should tell the same physical story, as long as you interpret it correctly. Make sure you understand this!

Show your work, of course.

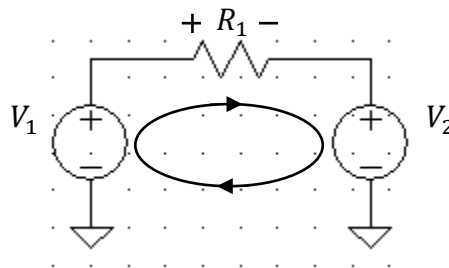
Problem 3: Ohm's Law (Reprise):

Repeat the previous problem with these values: $V_1 = 3.3V$, $V_2 = 5.0V$, and $R_1 = 500 \Omega$. First write the solution algebraically, then substitute numerical values for the variables. What does the sign of the current tell you? Which way is current flowing?

Topic: Review Kirchhoff's Laws:

G. Kirchhoff (1824-1887) formulated two laws in 1845 that are quite useful for analyzing networks (circuits). This should be a review of the material covered in Physics 14.

Kirchhoff's Voltage Law (Loop Law): The sum of voltages around a loop must equal zero.



Generally: $\sum_{loop}(V_i) = 0$
 In this case: $\sum_{loop}(V_i) = +V_1 - V_{R1} - V_2 = 0$

A helpful analogy is to say a change in voltage is akin to a change in elevation; both create a change in potential (electrical, gravitational). You have more potential 50V above a baseline than you do at 10V above that same baseline, much in the same way that you have more potential energy at 100m elevation above a baseline than you do 10m above. Higher levels = more potential in both cases. For example, imagine you were to hike Mount Moosilauke.

1. You begin at the Ravine Lodge, which we define as elevation, $P = 0\text{m}$.
2. If you follow the Gorge Brook Trail, you'll climb to about $P = +780\text{m}$. You've gone up in elevation and thus, potential.
3. You stop half way down at $P = +400\text{m}$ for a rest. You're still at a higher potential than you were at the Ravine Lodge, but now you have less potential than you did at the summit.
4. You complete the rest of your journey and return to the Lodge, at $P = 0$.

What was the net change in elevation? We've completed a loop from our starting point to our end point.

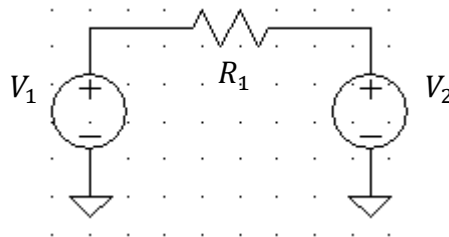
$$\sum_{\text{loop}} P = +780 - (780 - 400) - 400 = 0\text{m}.$$

Our net change in elevation is zero around the loop, even though we climbed to a high elevation!

Elevation data found at Hiking Mount Moosilauke, New Hampshire. home.earthlink.net/~ellozy/mt-moosilauke.html.

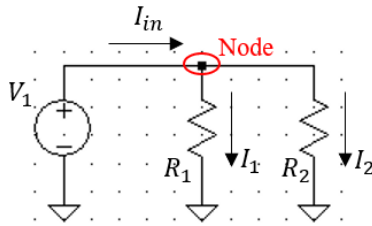
Problem 4: Kirchhoff's Voltage Law:

- (a) Define your polarities on the diagram.
- (b) Draw a loop to show where you plan to take KVL:



- (c) Solve for V_{R1} in terms of V_1 and V_2 , where $V_1 = 770\text{V}$ and $V_2 = 400\text{V}$. Show your work. Derive an algebraic expression for the solution, then substitute numerical values.
- (d) If you wanted I_{R1} to be no more than 1A, how big does R_1 need to be? Show your work. Derive an algebraic expression for the solution, then substitute numerical values.

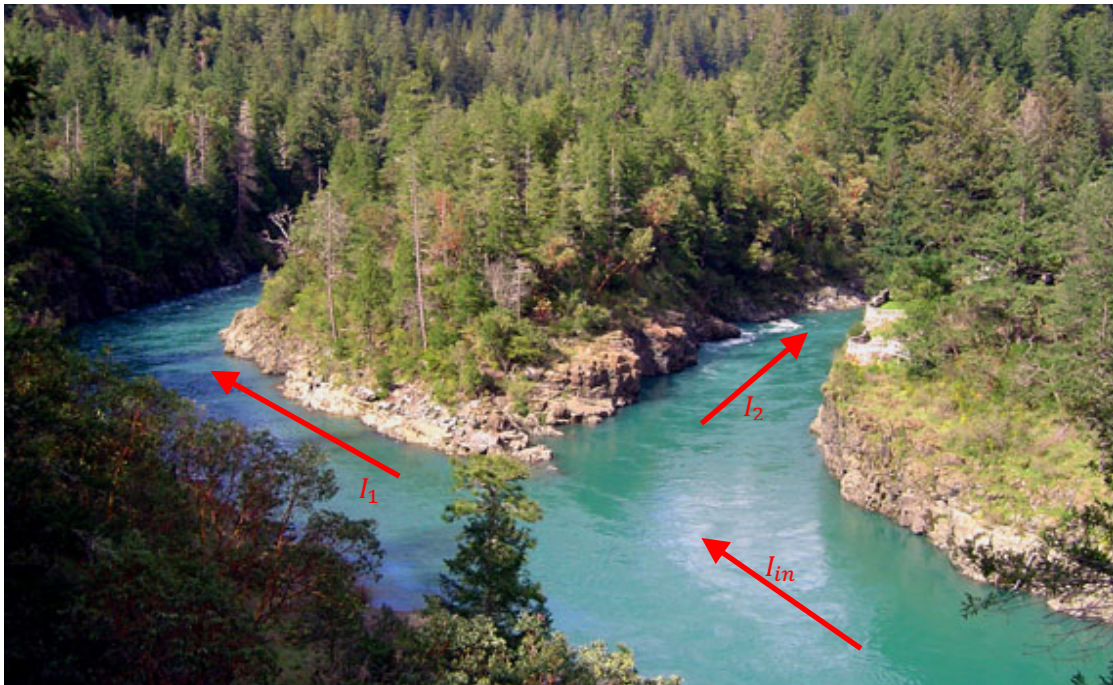
Kirchhoff's Current Law (Node Law): The sum of currents flowing into and out of a node must be equal to zero; charge is conserved.



Generally, $\sum_{node} I = 0$
 For this example, $\sum_{node} I = +I_{in} - I_1 - I_2$

A node is a place where circuit components meet. In the above example, V_1 , R_1 , and R_2 are all connected to the same node (circled in red). Currents flowing into the node are defined as positive and currents flowing out of the node are negative (I_{in} is flowing into the node, I_1 and I_2 are flowing out of it).

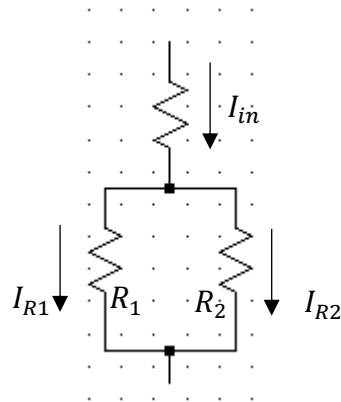
Think of electrical current as analogous to water flowing in a river. Water never spontaneously disappears—it travels into one branch or the other as the river forks.



Source: "Two Branches, One River." Two Branches, One River - Uncategorized - - View from the Sparrow's Nest, viewfromthesparrowsnest.com/?p=577.

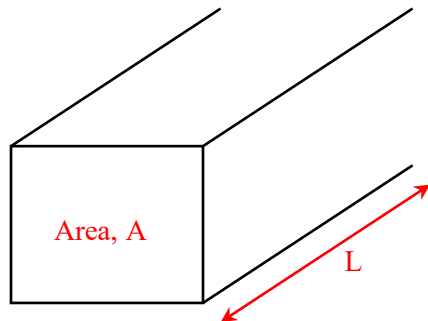
Problem 5: Kirchhoff's Current Law:

- (a) Define your polarities on the diagram based on how currents are defined.
- (b) Circle the node where you plan to take KCL.



- (c) Solve for I_{R1} in terms of I_{in} and I_{R2} , where $I_{in} = 3A$ and $I_{R2} = 1.2A$. Show your work. Derive an algebraic expression first, then substitute numerical values.
- (d) If $R_1 = 500\Omega$, what is the voltage drop across the resistor? Show your work. Derive an algebraic expression first, then substitute numerical values.
- (e) Solve for V_{R2} with KVL. Draw the loop on the diagram and work through the math. Show your work. Derive an algebraic expression first, then substitute numerical values.

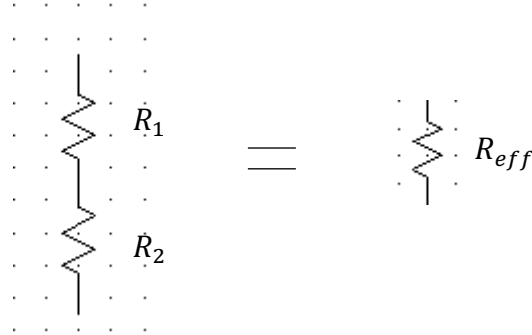
Topic: Resistance.



The resistance (ohms) of an object is given by $R = \frac{\rho L}{A}$ where L and A are defined above and ρ is the resistivity in ohms-meter (not a real intuitive dimension, but it works out). Increasing the length of the object increases its resistance, while increasing the cross-sectional area decreases its resistance. This is intuitive mathematically from the above equation. It's worth your time to think about why this is, physically, discussing with your partner.

Resistors in Series:

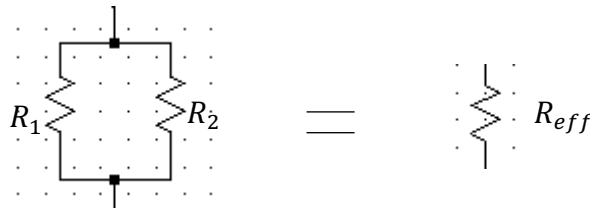
If two resistors are placed in series their effective resistance increases because the length is increased.



$$\text{Generally, } R_{eff} = R_1 + R_2$$

Resistors in Parallel:

If two resistors are placed in series their effective resistance decreases because the cross-sectional area is increased.



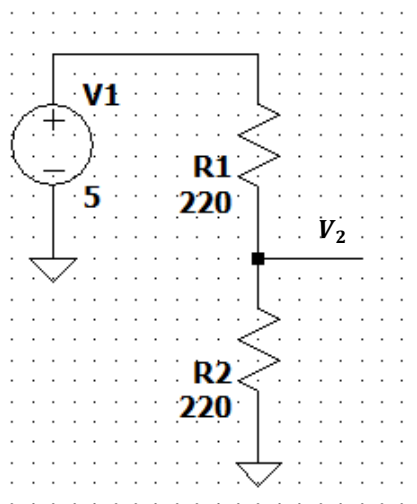
$$\text{Generally, } R_{eff} = \frac{R_1 R_2}{R_1 + R_2}$$

These expressions can be derived using Kirchoff's voltage and current laws.

Topic: Voltage Dividers and Current Dividers:

Armed with knowledge of Ohm's Law and Kirchhoff's Laws, you will analyze a circuit that is quite useful: the voltage divider.

Problem 7: Voltage Divider Circuit:



- (a) Draw polarities on the schematic and current directions on the diagram.
- (b) Solve for the voltage at the node between the two resistors both symbolically and numerically in terms of V_1 , R_1 , and R_2 . Do not set $R_1 = R_2$ and cancel in your symbolic solution—keep them independent (we often have voltage dividers with $R_1 \neq R_2$). What should R_1 and R_2 be in order to make $V_2 = 3.5V$?

As always, show your work. Derive an algebraic expression first, then substitute numerical values for the variables.

Topic: Diodes:

A diode is a circuit element that allows current to flow in only one direction, from a positive *anode* terminal to a negative *cathode* terminal.

Some, but not all, diodes emit light when current passes through them. These are called, appropriately, light-emitting diodes, abbreviated “LED”. LEDs are widely used as indicator lights in electronics, headlights and taillights for motor vehicles, in arrays for large image displays, and for general indoor and outdoor lighting applications. They are rugged and consume far less power than incandescent or fluorescent lights.

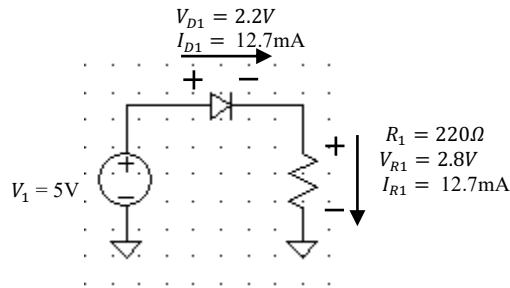
The circuit symbol for a diode shows the anode as a triangle, and the cathode as a flat bar (see the list of symbols earlier this is handout). LEDs that you might plug into a breadboard often have a longer lead to denote the anode and/or a flat spot on the side of the case to denote the cathode.

When a diode is on, it has a fixed voltage drop, V_F . For an LED, the value of V_F depends on the color of the LED. This is a function of the material properties of the diode.

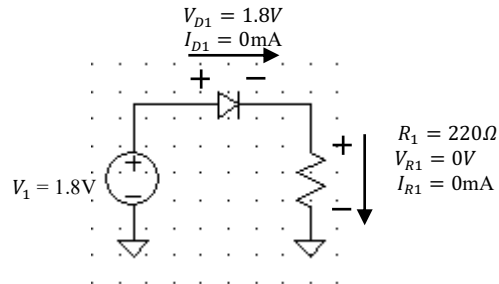
Typical LED Characteristics			
Semiconductor Material	Wavelength	Colour	$V_F @ 20mA$
GaAs	850-940nm	Infra-Red	1.2v
GaAsP	630-660nm	Red	1.8v
GaAsP	605-620nm	Amber	2.0v
GaAsP:N	585-595nm	Yellow	2.2v
AlGaP	550-570nm	Green	3.5v
SiC	430-505nm	Blue	3.6v
GaN	450nm	White	4.0v

Source: “Light Emitting Diode or the LED Tutorial.” *Basic Electronics Tutorials*, 24 Feb. 2018, www.electronicstutorials.ws/diode/diode_8.html.

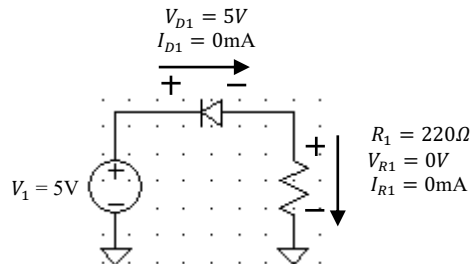
A diode turns on when the applied voltage is greater than the forward voltage (voltage from the anode to the cathode is positive) and enough current can flow. In the circuit below, assume that $V_1 = 5V$, $V_{F,D1} = 2.2V$ and $R_1 = 220\Omega$. This LED will be on because the voltage applied from the anode to the cathode is large enough and the resistance is small enough that current can flow. Drawing too much current will damage the LED. Always check the datasheet to see the maximum current that the device can draw and pick a current limiting resistor so the current is below that threshold.



Now assume that for the same schematic, $V_1 = 1.8V$, $V_{F,D1} = 2.2V$ and $R_1 = 220\Omega$. This LED will not turn on because there is not enough voltage in this system to create a 2.2V drop across D1.



When the diode is reversed, current is unable to flow from the source of potential and thus the LED does not turn on. Assume $V_1 = 5V$, $V_{F,D1} = 2.2V$ and $R_1 = 220\Omega$.



You will do a calculation with this result in the lab.

Before attempting Lab 1, discuss any questions you have with a TA or professor.